# Codemized - Practical Implementation Exercise

## Introduction

This exercise consists of a **practical implementation** designed to evaluate your ability to build a functional full-stack application.

You may use **AI tools (such as ChatGPT, Copilot, Cursor, or others)** while completing the exercise. However, it is important that you **understand the architecture and technical decisions** behind what you build, since during the review we may discuss how your solution works.

The goal is not to evaluate memorization or typing speed, but rather your ability to **design, structure, and deliver a functional and well-organized solution**.

**The exercise can be published directly in your GitHub account and you may keep it public so it can also become part of your portfolio for future interviews, if you wish.**

The objective of this exercise is to evaluate the following skills:

- Data modeling capability
- Code organization and separation of responsibilities
- Full-stack implementation
- Basic persistence handling
- Overall solution quality
- Ease of installation and project execution

## Assignment

You must implement an application that manages **users, projects, and tasks**.

The minimum entities to implement are the following:

**User**

Represents the person using the application. A user must be able to register, log in, and operate within the system as an authenticated user. Additionally, a user may create projects and be responsible for tasks.

**Project**

Represents a work unit or main grouping element within the application. Each project must be associated with a creator user and should function as a container for related tasks.

**Task**

Represents a specific unit of work within a project. Each task must belong to a project and may be assigned to a responsible user.

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**Comment**

Represents a discussion or update associated with a task. Comments allow users to collaborate by adding notes, clarifications, or progress updates related to a specific task.

The minimum relationships expected are:

- A **user** can have multiple **projects**.
- A **project** belongs to a **creator user**.
- A **project** can contain multiple **tasks**.
- A **task** belongs to a **project**.
- A **task** can be assigned to a **user**.
- A **task** can contain multiple **comments**.
- A **comment** belongs to a **task** and is created by a **user**.

The application must also include a **functional login system**, so that user authentication is required to operate within the platform.

The application should allow at least the following operations:

- User registration and login
- Project creation
- Listing projects belonging to the authenticated user
- Creating tasks within a project
- Listing tasks by project
- Assigning a task to a user

## Technical Guidelines

The challenge consists of developing a **full-stack application** using **Next.js** for the frontend. For the backend, you may use the framework of your choice (although we recommend Spring Boot or Nest.js) as long as the implementation follows a layered architecture, including at least a clear separation between **controller**, **service**, and **repository**. For the frontend, we require a proper componentization, styling practices, and efficient lifecycle and rendering management in the frontend.

**Backend-as-a-Service solutions such as Supabase will not be accepted.**

In addition to the functional implementation, the exercise must include everything necessary to run the application locally. The repository must be publicly cloneable and the application should start easily using **Docker Compose**, by running the command:

```
docker compose up
```

This command should initialize the backend and frontend along with all required dependencies, such as databases, caches, or other services.

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The repository is expected to include the scripts, configuration, and files required so that any reviewer can run the project without complex manual steps.

A **README.md** file must also be included describing the architecture of the project and any other considerations necessary to properly initialize it in a local environment.

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